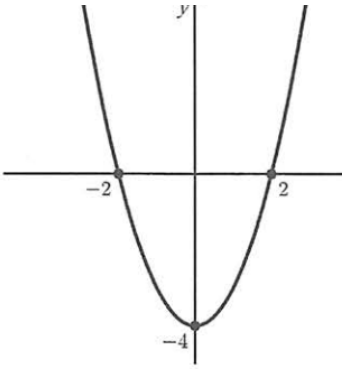
Graph Sketching: Reciprocal Functions

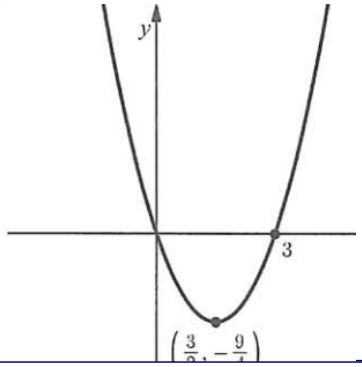
Questions 1–9

Question 1. For each of the following curves $y = f(x)$ shown below, sketch $y = \dfrac{1}{f(x)}$.

a)



b)



Question 2. Sketch each of the following, labelling any local maxima or minima with their coordinates:

a) $y = \dfrac{1}{3x - 1}$

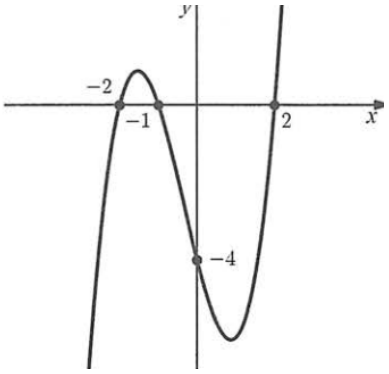
b) $y = \dfrac{1}{(x - 4)(x + 3)}$

c) $y = \dfrac{1}{2x^2 + 3x - 2}$

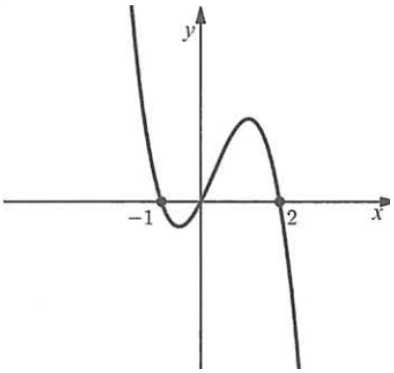
d) $y = \dfrac{1}{4 + 3x - x^2}$

Question 3. For each of the following curves $y = f(x)$ shown below, sketch $y = \dfrac{1}{f(x)}$.

a)



b)



c)

Question 4. Sketch each of the following:

a) $y = \dfrac{1}{x(x-1)(x-2)}$

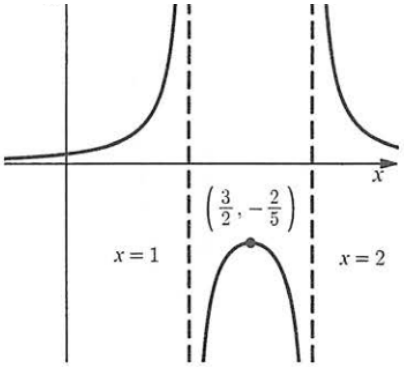
b) $y = \dfrac{1}{2x^3 - 9x^2 + 3x + 4}$

c) $y = \dfrac{1}{8 - 2x - 5x^2 - x^3}$

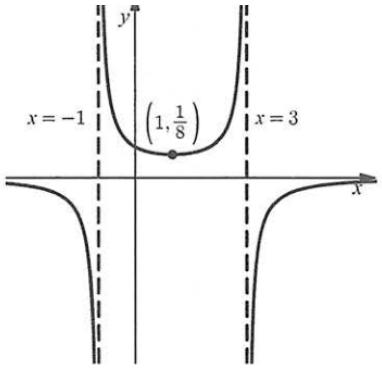
d) $y = \dfrac{1}{x^3 + x^2}$

Question 5. Each of the following curves are of the form $y = \dfrac{1}{ax^2 + bx + c}$, where a, b and c are integers. Determine the values of a, b and c.

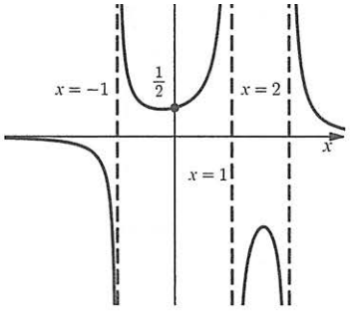
a)



b)



Question 6. The following curve is of the form $y = \dfrac{1}{ax^3 + bx^2 + cx + d}$, where a, b, c and d are integers. Determine the values of a, b, c and d.



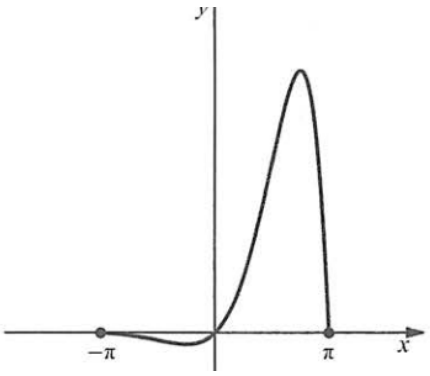
ketch the graph of $y = \frac{1}{f(x)}$ where

Question 7. Sketch the graph of $y = \dfrac{1}{f(x)}$, where

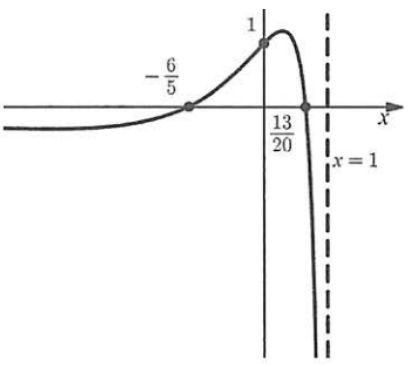
$$f(x) = \frac{x^2 + 4x}{x^2 - 4x - 5}.$$

Question 8. For each of the following curves $y = f(x)$ shown below, sketch $y = \dfrac{1}{f(x)}$.

a)



b)



Question 9. The following curve is of the form

$$y = \frac{1}{ax\ln(bx)},$$

where a and b are non-zero integer constants to be found. The point P is at $\left(\frac{1}{2e}, -\frac{e}{4}\right)$. There are asymptotes at $x = 0$ and $x = \frac{1}{2}$. Determine the values of a and b.

